

Supplementary Material

Supplementary Note S1: Interactive map

Supplementary File S1:

Interactive map of the full dipwell monitoring network (Google Maps, accessible at :https://www.google.com/maps/d/viewer?mid=1hXLAauiMeaVsXhBR_IoUTziAtjk)

Supplementary Note S2: Aquifer Geometry, Hydraulic Parameterisation and the Basis of the Spatial Flow Model

Hollingham (2026) — Newborough Warren Coastal Sand Dune Aquifer, Isle of Anglesey

2.1 The Borehole Data Problem

Darcy's law and any spatially explicit groundwater flow model require three spatially resolved inputs: hydraulic conductivity (K), aquifer thickness (b , from which transmissivity $T = K \cdot b$ is derived), and hydraulic gradient (∇h). For the Newborough Warren dune system, only the third of these — the hydraulic gradient — is well constrained, through the dipwell network of over 70 wells. Hydraulic conductivity and aquifer thickness are poorly constrained, because borehole logs providing lithological or geophysical evidence of the depth to the bedrock or glacial till substrate are absent across the great majority of the 700 ha dune system.

The available borehole evidence is confined to four locations documented in Betson et al. (2002), a report commissioned by the Countryside Council for Wales (Contract FC†73-05-18) covering hydrogeological investigations at the site. These boreholes provide minimum depth-to-basement constraints at four spatial points and constitute the only direct subsurface lithological evidence available for the site. They are summarised in Table S1.

Borehole	Easting (OSGB36)	Northing (OSGB36)	Saturated thickness (m)	Source
Water borehole	241,721	364,133	12.8 (min.)	Betson et al. (2002)
NH1	241,734	363,306	6.5	Betson et al. (2002)
NH2	242,024	363,107	6.5	Betson et al. (2002)
Borehole 3	242,837	363,288	3.65	Betson et al. (2002)

Table S1. Hard borehole constraints used as fixed nodes in the aquifer thickness IDW surface. Values represent confirmed minimum depth to the low-permeability substrate (glacial till or bedrock). The eastern borehole (Borehole 3) shows progressive thinning toward the Menai Strait.

The tracer test reported by Betson et al. (2002) provides the single direct estimate of hydraulic conductivity for the site, yielding $K = 6.0$ m/day at the central dune plain (range 3.0–9.0 m/day from the sensitivity analysis). This value is used unchanged as the site-wide K constant in the spatial model, with the range providing the basis for the exploratory uncertainty envelope noted in the figure annotations.

S2.2. Deriving the Aquifer Thickness Surface

S2.2.1. The β_1 Proxy

The four borehole constraints are insufficient to interpolate a site-wide thickness surface from direct evidence alone. The principal additional constraint used here is an indirect one, derived from the SSM β_1 (recharge sensitivity) coefficient.

The physical interpretation of β_1 is the fraction of a unit rainfall pulse that reaches the water table as a head rise, integrated over the monthly timestep. In an unconfined aquifer, this is governed by the depth of the unsaturated zone and the specific yield: a thick unsaturated zone above a large storage volume attenuates the rainfall-to-head transfer, producing a low β_1 . A shallow, low-storage aquifer (thin unsaturated zone, low S_y) produces a high β_1 — rainfall arrives at the water table quickly and raises it sharply. Although β_1 cannot be inverted algebraically to yield aquifer thickness without independent knowledge of the recharge flux and specific yield, its spatial pattern is a reliable proxy for the relative contrast in aquifer thickness across the site.

The Eastern Block (C1: $\beta_1 = 1.565$; C2: $\beta_1 = 1.674$) and Western Block (C3: $\beta_1 = 1.201$) contrast is consistent with the geological interpretation of Stratford et al. (2007): the Eastern Block sits on shallow till and estuarine deposits, producing a thin, storage-limited aquifer with a flashy response; the Western Block occupies deep, clean aeolian sand, producing a capacious, buffered aquifer with an attenuated response. The Forest cluster (C4: $\beta_1 = 0.957$) is anomalously low even relative to C3, reflecting canopy interception rather than additional aquifer depth — confirmed by the depth-dependent PET analysis (script 15) which found $\lambda = 0$ optimal for C4, indicating that capillary disconnection at depth is not the primary control.

Cluster	Label	β_1 (median)	Thickness prior (m)	Basis
C1	Eastern lake-buffer	1.565	7.0	NH1, NH2 boreholes; shallow till substrate
C2	Eastern mature dune	1.674	7.0	NH1, NH2 boreholes; consistent with C1
C3	Western mature dune	1.201	16.0	β_1 proxy (deep sand); Betson et al. (2002) western section; DEM ridge geometry
C4	Forest	0.957	16.0	Same deep sand substrate as C3; confirmed by Pearson affinity persistence post-felling
C5	Coastal	1.047	4.0	Geomorphological thinning toward Menai Strait
C6	Lake	—	0.5	Near-zero: open water at Llyn Rhos Ddu

Table S2. Cluster thickness priors used in the IDW thickness surface. C3 and C4 share the same prior because the Forest plantation sits on the same deep aeolian sand body as the Western Block, confirmed by β_1 similarity and Pearson affinity analysis.

S2.2.2. Per-Well Overrides and Boundary Pinch-out

CEH14 (241,292 E, 364,488 N), classified as C4, sits at the crest of the bedrock ridge at 14.4 m AOD with a mean water table head of 13.3 m AOD. The water table is only approximately 1.1 m below ground surface here. Given the ridge geology (fractured bedrock with thin sand veneer), the saturated thickness is estimated at 5 m rather than the C4 prior of 16 m.

Wells CEH7 (243,386 E, 363,613 N) and CEH8 (243,150 E, 363,382 N) sit at the far eastern estuarine margin of the site, where the DEM confirms ground elevation of 1.2–5.2 m AOD. Both are treated as pinch-out nodes with thickness 1.0 m. Shore boundary nodes (0.5 m) are placed along the southern coast, eastern coast, and at Llyn Rhos Ddu.

S2.2.3 IDW Interpolation and Smoothing

All hard borehole nodes and per-well prior/override values are combined into a single point set and interpolated to the 50 m grid using inverse-distance weighting (IDW) with power = 1 (linear distance weighting). Power = 2 was tested first but produced bullseye artefacts around the isolated borehole nodes. Power = 1 gives a smoother surface that interpolates plausibly between the sparse hard constraints and the denser well-level priors. A physical minimum of 0.3 m is enforced. The resulting thickness surface ranges from 0.3 m at the shore margins to approximately 18 m in the western interior.

S2.3. Why C4 Retains Its Cluster Identity After Clearfell

A key question for the scenario analysis is whether the clearfell treatment wells (FE1, FE2, FE4, LIS1) converged toward C3 open-dune behaviour after the December 2017 felling. The Pearson affinity analysis (Section 4.3) provides a direct empirical test: all four impact wells maintained $r > 0.97$ affinity with C4 even in the post-felling period (2018–2026), with no significant shift toward C3 affinity.

This persistence is consistent with the interpretation that the Forest cluster signature at these wells reflects deep sandy substrate at high topographic elevation rather than canopy influence alone — in which case convergence toward open-dune behaviour may not be achievable regardless of management intervention. The third interpretation is consistent with the β_1 proxy evidence: C4 wells sit at the highest topographic positions in the dune system (mean head 9.52 m AOD), and their deep unsaturated zones and low recharge sensitivity reflect their geological position, not only their canopy cover. This conclusion is independently supported by the NW10 broadleaf comparison (Section 4.6.6): NW10, situated within the 1993 clearfell and 1996 broadleaf restocking block, has maintained C4 cluster affinity ($r = 0.986$) throughout the 18-year monitoring period despite its different canopy cover, confirming that substrate position rather than canopy type is the dominant control on cluster identity.

S2.4. Summary of Model Parameterisation

Parameter	Value	Source / basis
Hydraulic conductivity K	6.0 m/day (range 3–9)	Betson et al. (2002) tracer test, central dune plain
Aquifer thickness (hard constraints)	3.65–12.8 m	Betson et al. (2002) borehole logs (4 locations)
Aquifer thickness (C1/C2 prior)	7.0 m	NH1/NH2 borehole evidence; Eastern Block β_1 contrast
Aquifer thickness (C3/C4 prior)	16.0 m	β_1 proxy (deep sand); Betson et al. (2002) western section
CEH14 override	5.0 m	Ridge crest geology; WT only 1.1 m below ground
CEH7/CEH8 override	1.0 m	Estuarine margin pinch-out; DEM ground

		elevation 1–5 m AOD
Grid resolution	50 m	Balances spatial detail against borehole data density
IDW power	1 (linear)	Power = 2 produced bullseye artefacts at sparse borehole nodes
Sea boundary (south/east)	$h = 0$ m AOD	Dirichlet BC; Menai Strait / Caernarfon Bay at OD
Sea boundary (west cap)	$N \leq 363,400$	DEM confirms ridge (>15 m AOD) above this Northing on western margin

Table S3. Summary of spatial model parameterisation. All values are directly traceable to borehole evidence, published tracer tests, or the SSM β_1 proxy analysis described in this note.

The principal uncertainty in the spatial model is the aquifer thickness surface in the western interior, where the C3 prior of 16 m is derived indirectly from the β_1 proxy and the Betson et al. (2002) western section rather than from direct borehole evidence. Slug tests or pumping tests at two to three representative C3 wells and a ground-penetrating radar survey of aquifer thickness are identified as the highest-priority data gaps for future field campaigns (Section 6.4).

Supplementary Note S3: Scenario Modelling Framework and Limitations

S3.1. Purpose and Scope of the Scenario Analysis

An exploratory scenario analysis was developed to investigate whether the site's own SSM β coefficients could yield defensible quantitative predictions of water table response to management interventions and climate change. This note documents the scenario framework adopted (the per-well equilibrium framework described in Section 3.8), the alternative PDE-based approach that was evaluated and rejected during development, and the structural limitations that apply to any equilibrium representation of the system. The scenario outputs themselves are reported in Section 4.9 and discussed in Section 5.6 of the main paper; this note provides the technical reasoning behind the methodological choices and their limitations.

S3.2. The Per-Well Equilibrium Framework

The scenario framework adopted in this study is a well-level equilibrium calculation that applies scenario-specific perturbations to the SSM forcing terms (rainfall, PET, canopy interception, atmospheric draw) and computes the resulting change in equilibrium head for each reference well. Per-well Δh values are then interpolated to a 50 m grid on the British National Grid by Delaunay triangulation with linear barycentric weighting (`scipy.interpolate.griddata, method="linear"`), masked to the rectangular sea-boundary extent; this produces the elevation-based maps presented in Figures 31 and 32. The accompanying interactive scenario viewer (`scenario_viewer.html`) renders the same per-well Δh field using power-1 eight-nearest-neighbour inverse-distance weighting for responsiveness under slider-driven re-interpolation, and additionally offers a depth-below-surface view (DEM minus interpolated head, per cell) with dune-ridge cells masked — see S3.7 for the viewer-specific rendering conventions. The full per-scenario $\times$ per-season $\times$ per-cluster Δh output is tabulated in the supplementary file `19_scenario_summary.csv`.

The scenario head change for each well is computed directly from the fitted SSM equation as

$$\Delta h = (\beta_1 \cdot P_{\text{eff,sc}} - \beta_2 \cdot \text{PET}_{\text{sc}} - \beta_3 \cdot |\bar{h}|) - (\beta_1 \cdot P_{\text{eff,0}} - \beta_2 \cdot \text{PET}_0 - \beta_3 \cdot |\bar{h}|)$$

where the $\beta_3 \cdot |\bar{h}|$ term cancels algebraically between scenario and baseline expressions, so β_3 does not materially affect the computed Δh . Specific yield does not appear in the equation because the fitted β coefficients already embed the aquifer storage response through the SSM fit to observed head data; the per-well WTF-derived Sy values (Section 3.7.3) are retained as supplementary display information in the interactive viewer, not as a divisor on Δh . The framework is honest about what it is: a geometric aggregator of SSM responses across the cluster structure, not a physical model of groundwater flow.

The per-well framework has two important structural features that shape the interpretation of its outputs. First, scenario parameter changes for canopy interception and β_2 apply only to the C4 Forest cluster wells, because that is where the relevant physical changes occur under forest management. The framework therefore necessarily produces a response confined to the C4 zone and its immediate interpolated surroundings, with effectively no propagation into C1, C2, or wider C3 clusters. Second, climate scenario parameter changes (ΔP , ΔPET) apply to all reference wells simultaneously, so the resulting Δh field is spatially continuous and reflects the site-wide β coefficient structure. The contrast between forest management and climate scenarios in the spatial outputs is therefore a direct consequence of which wells are affected by which perturbations, not an artefact of the numerical method.

S3.3. Evaluated-and-Rejected: The 2D Steady-State PDE Model

During development, an alternative spatial framework was implemented as a two-dimensional finite-difference PDE solver based on the Helmholtz form of the steady-state groundwater flow equation:

$$\nabla \cdot (T \cdot \nabla h) - \beta_3 \cdot h = -(\beta_1 \cdot P_{\text{eff}} - \beta_2 \cdot PET)$$

with transmissivity $T = K \cdot b$ ($K = 6.0$ m/day after Betson et al., 2002; b IDW-interpolated from four borehole constraints, also after Betson et al., 2002). Dirichlet head conditions ($h = 0$ m AOD) were applied at sea boundaries, and implicit Neumann no-flow conditions at the ridge. The sparse linear system of approximately 4,200 equations was solved using `scipy.sparse.linalg.spsolve` (Virtanen et al., 2020).

This approach is well-posed mathematically — the Helmholtz structure guarantees a unique solution with the given boundary conditions — and was physically consistent in that the β coefficients entering the source term are the same parameters estimated from the observed well records. The solver produced physically plausible baseline head surfaces and Darcy flux fields qualitatively similar to those obtained from the per-well framework.

However, the PDE solver produced near-zero site-wide differences for forest management scenarios (mean $\Delta h = -0.003$ to -0.008 m for full clearfell and thinning), for three structural reasons.

First, the forest management scenarios modify β coefficients only for the nine C4 Forest wells. These per-well β values are then IDW-interpolated to the 4,235-cell grid. Because the nine C4 wells constitute a small minority of the network of over 70 wells, the modified β values are heavily diluted by the surrounding clusters during interpolation. The source term $\beta_1 \cdot P_{\text{eff}} - \beta_2 \cdot PET$ therefore changes by only a small amount at each cell, and the resulting change in the steady-state head solution is proportionally small.

Second, the internal drainage term $\beta_3 \cdot h$ provides a stabilising feedback in the Helmholtz equation. C4 has the lowest β_3 in the network (0.078, compared with 0.118 for C3 and 0.200 for C2). This means that in the C4 zone the drainage feedback is weak, and a moderate source term change produces only a small equilibrium head change.

Third, and most fundamentally, the steady-state PDE does not capture the ridge-derived boundary subsidy that sustains the C4 water table. The observed mean head at C4 (9.52 m AOD) is substantially higher than the mean at C3 (6.09 m AOD), not because of C4's β coefficients but because the C4 wells are proximal to the bedrock ridge and receive lateral recharge from the ridge slope. CEH14, the well

most proximal to the ridge, carries the largest persistent lateral subsidy in the network ($\alpha = +0.222$ m/month, Model B intercept audit; Section 4.4.2). The steady-state PDE source terms are parameterised from the SSM β coefficients alone and do not include this ridge contribution. As a result, the PDE solution underestimates the true C4 water table under both baseline and scenario conditions by a systematic amount related to the ridge contribution, and the difference between the two solutions is correspondingly small.

A fully calibrated continuous-flow model capable of representing the ridge boundary subsidy dynamically would require, at minimum: (i) slug tests at two to three representative wells per cluster to constrain the spatial distribution of K (currently constrained only by the single tracer test of Betson et al., 2002); (ii) a ground-penetrating radar survey of aquifer thickness to replace the β_1 -proxy thickness priors with direct measurements; and (iii) instrumentation of the ridge boundary subsidy pathway sufficient to parameterise a time-varying boundary flux term. These are the same parameter-sparsity constraints that produced calibration difficulties in the Betson et al. (2002) MODFLOW model at this site.

For these reasons, the PDE approach was judged to overclaim relative to the data available and was rejected in favour of the per-well equilibrium framework. The framework represents what the SSM parameterisation supports without claiming spatial flow dynamics the monitoring data cannot constrain.

S3.4. Climate Scenarios in the Per-Well Framework

Climate scenarios applied under the per-well framework are based on the UKCP18 Regional 12 km projections for Wales (Met Office, 2018) and the CHES-SCAPE bias-corrected projections of Robinson et al. (2023). Two scenarios are carried forward into the Results (Section 4.9): UKCP18 2050s (2040–2069) applies a winter precipitation increase of +10%, summer precipitation decrease of –15%, winter PET increase of +5% and summer PET increase of +20%; UKCP18 2080s (2070–2099) applies +20%, –30%, +10% and +35% respectively for the same four terms. Both scenarios use RCP8.5 central-estimate (50th-percentile) values. Perturbations are applied per season, with winter defined as November–March and summer as May–September; the climatological baselines against which the perturbations are applied are the 2005–2026 monthly means from the RAF Valley climate record. The resulting Δh for each well is computed for each season separately, and the annual Δh is reported as the 0.5-weighted mean of the winter and summer equilibrium responses — this convention captures the physically significant seasonal asymmetry between winter recharge and summer evaporative loss that an annual-mean forcing would obscure.

The framework predicts a site-wide annual mean equilibrium Δh of -0.040 m month⁻¹ under UKCP18 2050s (-0.48 m yr⁻¹ equivalent) and -0.072 m month⁻¹ under UKCP18 2080s (-0.86 m yr⁻¹ equivalent). Cluster-level annual responses range from -0.048 m month⁻¹ (C1 Eastern Block lake-buffer) to -0.012 m month⁻¹ (C5 Coastal) under the 2050s trajectory, and from -0.085 to -0.022 m month⁻¹ under the 2080s trajectory, preserving the $C1 > C2 > C3 > C4 > C5$ ranking of cluster climate-sensitivity established in the baseline β coefficient structure. The annual response conceals a pronounced seasonal asymmetry: winter Δh is modestly positive across all clusters (site-wide means $+0.008$ m month⁻¹ at 2050s, $+0.015$ m month⁻¹ at 2080s), while summer Δh is strongly negative and drives the annual signal (site-wide means -0.089 m month⁻¹ at 2050s, -0.159 m month⁻¹ at 2080s, reaching -0.192 m month⁻¹ at C1 under the 2080s trajectory). The positive winter response reflects the larger fractional increase in winter precipitation than winter PET under UKCP18 for Wales; the negative summer response reflects the combined effect of reduced summer precipitation and

substantially increased summer PET. The full per-scenario $\times$ per-season $\times$ per-cluster output is tabulated in 19_scenario_summary.csv (Section 4.9).

Three limitations on the interpretation of these outputs apply. First, the perturbations used are central-estimate values; the UKCP18 probabilistic range at the 10th and 90th percentiles spans roughly half to twice the central perturbation for precipitation and a broader range still for PET, so the Δh values reported here are a central point within a substantially wider envelope of plausible responses. Users of the interactive scenario viewer (scenario_viewer.html) can explore perturbations across the full 0.5–1.5 $\times$ range of the seasonal sliders to visualise this envelope. Second, the equilibrium framework computes the steady-state response to sustained seasonal perturbations and does not resolve within-year dynamical propagation of recharge between winter and summer — for example, the effect of a wetter winter on the summer minimum through carry-over storage is captured only to the extent that the SSM β_3 drainage coefficient encodes it in the fitted mean. Third, the framework holds the SSM β coefficients themselves fixed at their historical values; it does not allow for structural changes in the β coefficients under future climate (for example, a higher β_2 under consistently warmer summers through increased vapour pressure deficit). Within these limitations, the climate scenario output is the most quantitatively defensible use of the equilibrium framework: every cluster's response is derived from its own fitted β coefficients under a uniform climate perturbation that reflects the current best estimate for the site's regional climate trajectory, and the result reflects the physical partition of rainfall and PET sensitivity that the SSM has estimated from 21 years of observation.

S3.5. Forest Management Scenarios in the Per-Well Framework

Forest management scenarios (full clearfell, 50% thinning, broadleaf conversion) apply canopy parameter changes only to the nine C4 Forest wells, and the framework correctly produces a response confined to the C4 zone. This is a feature of the framework, not a bug: under the SSM parameterisation, canopy management directly affects only the wells beneath the canopy, and there is no structural mechanism for canopy effects to propagate into the open dune clusters beyond the hydraulic gradient already implicit in the baseline heads. The BACI monitoring record provides empirical corroboration of this structural feature: post-felling displacement is concentrated at the core impact wells, and wells outside the C4 cluster show no detectable felling-specific response once common-mode climate variability is accounted for (Section 4.6.4). Spatially-uniform deepening across the transect from clearfell core to uncleared plantation interior (CEH2 -0.111 m, CEH34 -0.093 m, WMC3 -0.140 m) reflects climate variability rather than propagation of the clearfell signal.

The clearfell scenario applies a β_2 multiplier of $\times 1.20$ at C4 wells, derived from the BACI-observed post-felling β_2 zone-mean ratios (impact: $3.755/3.124 = 1.20$; edge: $3.751/3.123 = 1.20$; control: $3.696/3.030 = 1.22$; Section 4.6.6). Canopy interception is set to 0%. The framework predicts $\Delta h_{C4} = -0.024$ m month $^{-1}$ annually (-0.29 m yr $^{-1}$ equivalent), with a strongly asymmetric seasonal profile (-0.050 m month $^{-1}$ in summer versus $+0.001$ m month $^{-1}$ in winter) reflecting the dominance of summer PET in the β_2 sacrificial shielding mechanism established in Sections 4.6.6 and 5.4.2. The BACI-observed raw summer displacement ($+0.292$ m relative to the full pre-clearfell baseline; Section 4.6.3) is substantially larger than the framework prediction, but the baseline for the BACI comparison incorporates the anomalously wet 2015–16 years and the 2015 scraping intervention, both of which inflate the apparent post-felling deepening (Section 4.6.4). Against a cleaner post-scraping pre-felling baseline the observed summer deepening is $+0.111$ m (Section 4.6.3), closer to the framework prediction. The framework output is presented as the equilibrium lower bound on the canopy-removal effect rather than as a full dynamical representation of the observed response.

The thinning scenario applies a β_2 multiplier of $\times 1.10$ (halving of the clearfell signal) with canopy interception reduced from 24% to 12%. The framework predicts $\Delta h_{C4} = -0.012 \text{ m month}^{-1}$ annually, approximately half the clearfell response as expected.

The broadleaf conversion scenario is represented by replacing the Corsican pine canopy interception fraction of 24% (Freeman, 2008) with an annual-mean deciduous interception of 15% (following the temperate deciduous meta-analysis of Komatsu et al., 2011) while holding β_2 at baseline. No annual-mean justification exists for a year-round β_2 adjustment under broadleaf: the canopy is leafed in summer and leafless in winter, and the annual mean β_2 is expected to be comparable to pine. The framework predicts a small positive $\Delta h_{C4} = +0.006 \text{ m month}^{-1}$ annually ($+0.07 \text{ m yr}^{-1}$ equivalent), reflecting the modest annual net recharge gain from reduced interception. This is a steady-state annual-mean approximation only; the seasonal asymmetry that underlies the phenological mechanism described in Section 5.4.4 — larger winter recharge pulse under leafless canopy, steepened hydraulic gradient, accelerated drainage into summer — is a dynamical trajectory response that the equilibrium framework does not resolve.

For the Figure 33 hydrograph (Section 5.6), a seasonally-varying β_2 is applied to the broadleaf scenario to represent the full-LAI summer canopy and leafless winter canopy. This seasonal parameterisation is a visualisation device and is not part of the Section 4.9 Results; it is used specifically to communicate the phenological argument from Section 5.4.4 alongside the BACI-observed record. The annual-mean output in `19_scenario_summary.csv` remains the quantitatively citeable framework result.

S3.6. Implications and Future Work

The per-well equilibrium framework is a lightweight analytical device for translating the SSM β coefficients into spatial summaries of equilibrium head response. Its strengths are that it makes no claims beyond what the SSM itself supports, that it is fully reproducible from the published CSV outputs, and that it is explorable through an interactive viewer that exposes all parameter choices to the user. Its limitations are those of any equilibrium framework: it does not resolve dynamical trajectory responses, it does not capture cluster-to-cluster hydraulic coupling, and it cannot simulate transient interventions in which the time evolution of the water table matters for ecological outcome. The framework is adequate for the questions the paper addresses — whether forest management in the C4 zone affects summer minima in C1 and C2, and how climate perturbations partition across the cluster structure — and the BACI monitoring record confirms the framework's structural conclusions.

Answering a different class of question, such as how a hypothetical intervention along the CEH14 ridge subsidy pathway would propagate through the aquifer, would require a calibrated continuous-flow model. For that, the critical unknowns are hydraulic conductivity (currently constrained by a single tracer test at $K = 6.0 \text{ m/day}$; Betson et al., 2002) and aquifer thickness (inferred from four borehole logs across 700 ha). Slug tests at two to three representative wells per cluster and a ground-penetrating radar survey of aquifer thickness are identified as the priority field measurements required to enable a properly calibrated continuous-flow model, should future work pursue intervention scenarios where lateral hydraulic propagation is central to the management question.

Summary of scenario modelling approach

Primary framework: Per-well equilibrium with triangulation-linear grid rendering (Sections 3.8, 4.9; `scenario_viewer.html`; `19_scenario_summary.csv`)

Evaluated and rejected: 2D steady-state Helmholtz PDE solver (Section 3.8 PDE paragraph; S3.3 above)

Visualisation device: Figure 33 synthetic hydrograph with seasonally-varying broadleaf β_2 (Section 5.6; script 21)

Empirical corroboration: BACI monitoring record (Section 4.6)

S3.7. Interactive Viewer: Rendering Conventions and Depth-Mode Visualisation

The companion interactive scenario viewer ([scenario_viewer.html](#); Hollingham, 2026b) renders the same per-well Δh field that underpins the published Figures 31 and 32 of the main paper, but uses a distinct grid-rendering method at the browser to preserve responsiveness under slider-driven re-interpolation. The viewer implements inverse-distance weighting with power 1 and eight nearest neighbours (10 m distance floor; 5 m exact-value shortcut) rather than the triangulation-linear method used for the static figures. Power-1 IDW with a fixed neighbour set produces a smoother surface than the naïve power-2 IDW previously used, removes the bullseye halo artefacts that arose around individual wells on the Δh surface, and is sufficiently close to the triangulation-linear rendering at well locations that the two methods agree to within the mapped colour-scale resolution across the interpolated interior. At well locations the viewer returns the well's observed value unchanged; outside the convex hull of the wells it extrapolates under the IDW kernel where the triangulation-linear method would return no data. This distinction is most visible near the south-eastern coastal margin, where the viewer renders a short extrapolated zone that the static figures correctly leave blank.

The viewer offers three complementary display modes. The first two — change in water table elevation (Δh) and absolute water table elevation (m AOD) — are elevation-based surfaces and match the quantity displayed in Figures 31 and 32 respectively. The third mode, depth below surface (m), is a viewer-specific visualisation not reproduced in the static figures. It is computed per grid cell as $d = h_{\text{DEM}}(E, N) - h_{\text{WT}}(E, N)$, where h_{DEM} is the bilinearly-interpolated LiDAR DEM elevation at cell centre and h_{WT} is the IDW-interpolated scenario water table elevation at the same cell. This per-cell subtraction is the quantity that mediates the ecological response: the Curreli et al. (2013) slack vegetation thresholds are defined in depth-below-surface terms (SD15b wet slack summer viability limit: 0.61 m; SD16 dry slack viability limit: 0.98 m), and rendering the scenario water table against the topographic surface permits direct visual identification of the slack positions where scenarios cross these thresholds. The viewer's depth-mode colour scale is anchored on these thresholds: light blue at the 0.10 m winter flooding limit, green at 0.61 m, orange at 0.98 m, and deep red at 1.50 m, with explicit black tick marks on the legend at the SD15b and SD16 threshold depths.

A dune-ridge mask is applied exclusively in depth-below-surface mode. The rationale is that the water table is a continuous surface in space, and its elevation relative to ordnance datum is well-defined at every point within the site regardless of whether the overlying ground is a ridge or a slack floor; elevation-based maps (modes one and two) therefore require no topographic masking. Depth below surface, however, is meaningful only where the ground is approximately at the well-plane elevation — at a dune ridge the interpolated water table sits several metres below ground, but this is a consequence of the ridge height, not of any hydrological feature, and rendering such cells would produce a misleading strong-red patch on the depth map that obscures the ecologically relevant slack-floor response. The viewer masks a cell in depth mode when the DEM elevation at that cell exceeds the IDW-interpolated well-plane elevation (computed by the same IDW kernel applied to well DEM ground elevations) by more than 1.0 m. This threshold is user-toggable from the viewer's control panel; disabling it reveals the unmasked depth field, which can be useful for diagnostic inspection but should not be used for ecological interpretation.

Additional viewer conventions of note: the basemap is a LiDAR-derived greyscale hillshade rendered at approximately 35% opacity underneath the scenario surface, matching the static figure hillshade convention of `map_utils.load_dem_hillshade` with azimuth 315°, altitude 35° and vertical exaggeration 3; climate perturbations can be explored across the full 0.5–1.5× range per season, extending substantially beyond the UKCP18 2050s and 2080s central-estimate presets to allow sensitivity analysis across the probabilistic envelope; and tooltips on any well display both the baseline and scenario-perturbed depth values alongside the absolute head and Δh , providing a direct visual link between the per-well framework output and the ecological depth-below-surface interpretation.

Supplementary Note S4: Specific Yield mapping

Table S4. Individual Well WTF Specific Yield Estimates — Newborough Warren 2005–

2026: Event-based median specific yield (S_y) derived from the water table fluctuation (WTF) method (Healy and Cook, 2002) for 67 reference wells and 19 extended network wells. n events = number of qualifying monthly rising-limb events (criteria: $\Delta h > 5$ mm, net recharge $P - PET > 10$ mm). Q_{25}/Q_{75} = interquartile range; wide IQR reflects month-to-month variability in event estimates rather than measurement error. Forest cluster (C4) wells marked Interception corrected = Yes have net recharge adjusted for 24% canopy interception (Freeman, 2008). Extended well cluster assignments derived from Pearson affinity sitewide audit (Section 4.3). *Excluded from IDW interpolation surface: CEH12 (bedrock ridge — WTF response reflects fractured rock); CEH15 (forest slack floor — slack topography dominates water table dynamics). Wells shown in red. Generated by 18_wtf_spatial.py.

Well	Cluster	Network	n events	S_y median	S_y Q25	S_y Q75	Interception corrected	Excluded from interpolation
CEH11	1	Reference	50	0.223	0.167	0.322		
CEH22	1	Reference	26	0.313	0.178	0.366		
CEH23	1	Reference	34	0.192	0.121	0.275		
CEH25	1	Reference	49	0.172	0.142	0.286		
CEH26	1	Reference	39	0.223	0.130	0.315		
CEH27	1	Reference	43	0.212	0.150	0.314		
CEH5	1	Reference	52	0.192	0.139	0.323		
CEH6	1	Reference	55	0.211	0.142	0.309		
P2	1	Extended	17	0.238	0.106	0.296		
CEH10	2	Reference	50	0.245	0.119	0.328		
CEH24	2	Reference	43	0.216	0.144	0.323		
CEH28	2	Reference	35	0.227	0.146	0.374		
CEH40	2	Reference	22	0.271	0.168	0.321		
CEH41	2	Reference	24	0.222	0.132	0.289		
D10	2	Reference	37	0.235	0.156	0.369		
D15	2	Reference	36	0.198	0.143	0.308		
D17	2	Reference	42	0.251	0.175	0.338		
D38	2	Reference	47	0.216	0.144	0.307		
D41	2	Reference	37	0.227	0.154	0.377		
D43	2	Reference	45	0.226	0.158	0.340		
D44	2	Reference	45	0.237	0.179	0.355		

Well	Cluster	Network	Events	Sy median	Sy Q25	Sy Q75	Interception corrected	Excluded from interpolation
D5	2	Reference	40	0.259	0.140	0.382		
D6	2	Reference	48	0.267	0.158	0.415		
D7	2	Reference	42	0.213	0.128	0.333		
D8	2	Reference	36	0.219	0.140	0.367		
D9	2	Reference	38	0.211	0.153	0.354		
L7	2	Reference	47	0.212	0.139	0.299		
NW 4B	2	Reference	42	0.212	0.147	0.320		
NW13	2	Reference	36	0.207	0.127	0.266		
NW3	2	Reference	47	0.254	0.160	0.359		
NW4	2	Reference	42	0.233	0.158	0.333		
T41A	2	Reference	43	0.201	0.136	0.338		
T41B	2	Reference	41	0.212	0.163	0.397		
T41C	2	Reference	40	0.253	0.167	0.365		
T41D	2	Reference	36	0.214	0.157	0.320		
WMC1	2	Reference	35	0.232	0.135	0.350		
WMC4	2	Reference	36	0.199	0.117	0.278		
CEH12	2	Extended	8	0.366	0.338	0.380		Yes*
CEH35	2	Extended	8	0.218	0.120	0.330		
CEH37	2	Extended	28	0.197	0.128	0.308		
CEH38	2	Extended	13	0.303	0.170	0.450		
CEH7	2	Extended	22	0.209	0.144	0.318		
CEH8	2	Extended	25	0.246	0.151	0.348		
PE	2	Extended	16	0.243	0.182	0.299		
PW	2	Extended	17	0.220	0.136	0.229		
CEH 1	3	Reference	49	0.232	0.141	0.306		
CEH16	3	Reference	31	0.250	0.161	0.342		
CEH17	3	Reference	25	0.282	0.161	0.389		
CEH18	3	Reference	38	0.260	0.150	0.354		

Well	Cluster	Network	Events	Sy median	Sy Q25	Sy Q75	Interception corrected	Excluded from interpolation
CEH19	3	Reference	24	0.360	0.180	0.391		
CEH21	3	Reference	24	0.307	0.166	0.378		
CEH31	3	Reference	26	0.275	0.205	0.392		
CEH36	3	Reference	28	0.290	0.162	0.361		
CEH39	3	Reference	27	0.263	0.152	0.378		
CEH4	3	Reference	39	0.243	0.166	0.339		
CEH42	3	Reference	27	0.254	0.142	0.359		
CEH9	3	Reference	40	0.275	0.166	0.389		
D25	3	Reference	35	0.264	0.189	0.355		
NW1	3	Reference	47	0.173	0.118	0.282		
NW11	3	Reference	47	0.203	0.129	0.270		
NW2	3	Reference	47	0.204	0.115	0.296		
NW5	3	Reference	48	0.254	0.144	0.337		
NW6	3	Reference	41	0.237	0.188	0.309		
NW7	3	Reference	42	0.265	0.180	0.358		
NW9	3	Reference	34	0.282	0.166	0.339		
WMC2	3	Reference	38	0.234	0.176	0.348		
WMC3	3	Reference	34	0.258	0.226	0.382		
FE3	3	Extended	19	0.263	0.165	0.332		
NW8	3	Extended	14	0.233	0.140	0.357		
NW8B	3	Extended	26	0.266	0.222	0.363		
P1	3	Extended	7	0.313	0.225	0.394		
CEH13	4	Reference	38	0.237	0.143	0.333	Yes	
CEH14	4	Reference	40	0.154	0.094	0.290	Yes	
CEH2	4	Reference	42	0.182	0.116	0.252	Yes	
CEH20	4	Reference	37	0.197	0.133	0.271	Yes	
CEH30	4	Reference	34	0.223	0.138	0.293	Yes	
CEH32	4	Reference	35	0.197	0.143	0.274	Yes	

Well	Cluster	Network	n events	Sy median	Sy Q25	Sy Q75	Interception corrected	Excluded from interpolation
		e						
CEH33	4	Reference	35	0.221	0.148	0.286	Yes	
CEH34	4	Reference	35	0.196	0.124	0.278	Yes	
NW10	4	Reference	42	0.210	0.111	0.284	Yes	
CEH15	4	Extended	16	0.141	0.101	0.364	Yes	Yes*
FE1	4	Extended	24	0.196	0.109	0.267	Yes	
FE2	4	Extended	24	0.217	0.128	0.285	Yes	
FE4	4	Extended	20	0.228	0.160	0.281	Yes	
LIS1	4	Extended	20	0.183	0.129	0.258	Yes	
NW12	4	Extended	9	0.211	0.161	0.293	Yes	

Supplementary Note S5: Residual-Lag Test of the Ridge-Recharge Attribution

S5.1. Purpose

The boundary-subsidy argument developed in Sections 4.4.2 and 5.3 attributes the persistent water-balance residual at forest-margin wells — most visibly the +0.222 m/month intercept at CEH14 — to lateral recharge derived from the northern rock ridge. This attribution rests primarily on two lines of evidence: the spatially structured pattern of positive Model B intercepts concentrated along the forest–dune boundary (Figure 15a), and the water balance closure argument that no other plausible flux can account for the monthly deficit. Neither of these is a direct test of the transport mechanism itself. The residual-lag analysis described here provides an independent, falsifiable test: if ridge-derived recharge is delivered as a time-varying pulse, then water travelling from the ridge to successive dune wells must arrive at those wells with travel times that increase as a function of distance. The time structure of the SSM residuals, cross-correlated against rainfall, should therefore carry a distance-dependent lag signature. This Note documents the test and reports the null result.

S5.2. Extended State-Space Model

The single-period SSM fitted in Section 3.4.3 expresses the water table increment as a function of contemporaneous rainfall:

$$\Delta h(t) = \alpha + \beta_1 \cdot P(t) - \beta_2 \cdot PET(t) - \beta_3 \cdot h(t-1) + \varepsilon(t)$$

A preliminary analysis (Script 22) demonstrated that the residuals $\varepsilon(t)$ from this formulation carry a generic lag-1 rainfall signal at every well on the site, irrespective of cluster or position. This is the expected monthly-timestep consequence of the vadose zone: recharge takes approximately one month to propagate to the water table, so monthly-averaged water-table response lags monthly-averaged rainfall by roughly one month. This generic signal dominates the cross-correlation function at every well and masks any ridge-specific structure that might be present at longer lags.

To remove this confound, the residual-lag test uses an extended model that explicitly includes both $P(t)$ and $P(t-1)$ as regressors:

$$\Delta h(t) = \alpha + \beta_{10} \cdot P(t) + \beta_{11} \cdot P(t-1) - \beta_2 \cdot PET(t) - \beta_3 \cdot h(t-1) + \varepsilon'(t)$$

The β_{10} and β_{11} coefficients absorb the vadose-zone response between them; any remaining lag structure in $\varepsilon'(t)$ is therefore post-vadose and is the candidate ridge-transport signal. Fitted coefficients confirm that the extended parameterisation is well-identified: β_{11} is statistically significant ($p < 0.05$) at 100% of wells in every cluster, and the lagged-response fraction $\beta_{11} / (\beta_{10} + \beta_{11})$ increases monotonically from 0.55 in C1 to 0.83 in C4 — consistent with the slower drainage of forest soils. Mean R^2 improves from 0.55 (single-period) to 0.69 (extended). The extended model is used solely for this diagnostic test; the β_1 , β_2 , β_3 and α values reported in the main text remain the authoritative parameterisation for all other analyses.

S5.3. Cross-Correlation and Pre-Whitening

For each well with at least 140 months of data ($n = 57$ after excluding CEH3, CEH4, CEH7, CEH8 and CEH37 for reasons noted below), the Pearson cross-correlation $r(\varepsilon'(t), P(t - N))$ was computed for lags $N = 0$ to 12 months. Prior to cross-correlation, both the rainfall and residual series were pre-whitened using a first-order autoregressive filter: $x'(t) = x(t) - \phi \cdot x(t-1)$, where ϕ is the AR(1) coefficient of the rainfall series (+0.210 at RAF Valley). Wells whose residuals retained AR(1) structure above $|\phi| = 0.2$ were additionally pre-whitened against their own residual autocorrelation before the common filter was applied. Statistical significance was evaluated using the Bartlett 95% confidence interval for a white-noise series, $|r| \geq 1.96/\sqrt{N}$, which for typical series lengths gives a threshold of approximately 0.15.

Wells excluded from the test: CEH3 (tidal boundary — outside the SSM operational domain), CEH4 (coastal erosion drift plus post-2017 clearfell drawdown confounding any lag signal), CEH7, CEH8, CEH37 (standard upstream exclusions carried over from Section 4.4.2). Wells beyond the 140-month minimum were retained without further filtering.

S5.4. Hypothesis Test

The test metric is the Spearman rank correlation between each well's peak-correlation lag N^* and its Euclidean distance from the ridge reference point at (E = 241750, N = 364500, OSGB36) — a representative coordinate for the nearest point of the northern rock ridge to the dune field. Under the ridge-transport hypothesis, Spearman ρ should be significantly positive.

Of 57 wells fitted, 42 had a peak cross-correlation that exceeded the Bartlett significance threshold. Across these 42 wells, 33 showed their peak at lag 2 months and the remaining 9 distributed between lags 3 and 12 with no systematic geographic pattern. The Spearman rank correlation between peak lag and ridge distance was $\rho = -0.15$ ($p = 0.33$); a Mann-Whitney test specifically comparing C4 ridge-adjacent wells against C2+C3 wells confirmed that C4 peak lags (mean 2.78 months) were not longer than those of the geologically distant dune-body wells (mean 3.34 months; $U = 118.5$, $p = 0.74$). The hypothesis of distance-dependent ridge transport is not supported.

The magnitude of coupling tells a different story. C4 ridge-adjacent wells show substantially stronger rainfall-residual coupling at lag 2 (CEH34 $r = +0.40$; CEH13, CEH2, CEH14 all $r \approx +0.25$) than similarly close but geologically separate C3 wells (NW1 $r = +0.07$, not significant; NW10 $r = +0.15$). The strength of rainfall coupling is elevated at ridge-adjacent wells; its timing is not distance-structured.

S5.5. Interpretation

The null distance-lag result does not rule out ridge-derived recharge. It rules out one specific form of it: event-driven, pulse-delivered transport in which monthly rainfall on the ridge propagates to the dune field with travel times that scale with distance. The alternative interpretation — that ridge recharge is delivered as a near-steady baseflow, sufficiently smoothed by its transit through fractured bedrock that its month-to-month variance falls below the detection threshold of monthly water-level records — is consistent with the null. A steady baseflow of this kind would manifest observationally as a constant positive contribution to the local water balance at wells along the flow path, which is precisely what the Model B intercept α already absorbs. The two mechanisms are therefore not distinguishable from water-level data alone.

The discussion of the boundary subsidy in Section 5.3 has been adjusted to reflect this constraint: the ridge contribution is retained as a mechanism consistent with the data, but its form is characterised as steady-state rather than event-driven. Definitive resolution of the transport mechanism would require either a ridge-crest rain gauge paired with bedrock piezometers, or direct geochemical tracer work on the ridge–dune flow path. Both are identified as priority further work in Section 11.

S5.6. Reproducibility

The residual-lag test is implemented in two standalone analysis scripts, *22_residual_lag_analysis.py* and *23_ridge_recharge_lag_test.py*, which are maintained separately from the main pipeline to preserve the reproducibility of the headline results from the canonical 01–21 sequence. Script 22 generates the Model B residuals and AR(1) diagnostics; Script 23 fits the extended model, computes the cross-correlations, and applies the Spearman test. Both scripts read the same intermediate data produced by the main pipeline (*01_wells_clean.csv*, *01_climate.csv*, *01_locations.csv*, *02_cluster_stats.csv*) and produce independent outputs under *22_residual_lag_analysis/* and *23_ridge_recharge_lag_test/* in the project outputs tree. A plain-text summary of the hypothesis test result is written to *23_05_hypothesis_test_summary.txt* at runtime.

Supplementary Note S6: Seasonal Climatology Diagnostic of the Residual

S6.1. Purpose

Supplementary Note S5 reported a null result for the event-driven ridge-transport hypothesis. This Note addresses the other leading candidate mechanism for the water-balance residual: that the Thornthwaite PET estimate underestimates the true summer atmospheric demand on the water table because it is temperature-only and does not capture net radiation, vapour pressure deficit, wind, or surface-condition effects. If that were the dominant explanation, the Model B residuals should carry a systematic summer-negative signature that the fitted β_2 has not absorbed. This Note tests for that signature, along with two subsidiary diagnostics that distinguish between the remaining candidate explanations (steady ridge baseflow, nonlinear recharge response, and residual model error).

S6.2. Diagnostics

Three complementary tests are computed from the Model B residuals $\varepsilon_B(t)$ produced by Script 22 for the same 57 wells that passed the ≥ 140 -month record-length filter and the site-extent exclusions detailed in S5.3.

(i) **Seasonal climatology.** For each well, the monthly climatology of the residual is computed as the mean of $\varepsilon_B(t)$ by calendar month over the record, giving a 12-point annual cycle. A sinusoidal fit

$$\varepsilon_B(m) = a_0 + a_1 \cdot \cos(2\pi m/12) + a_2 \cdot \sin(2\pi m/12)$$

yields an amplitude $\sqrt{a_1^2 + a_2^2}$ and a phase (month of peak). Summer-minus-winter magnitude (JJA mean – DJF mean) is reported as a robust non-parametric proxy for the same quantity.

(ii) **Independent ET proxy.** A correlation of the residual against PET itself would be trivially zero because OLS fitting makes residuals orthogonal to every regressor by construction. Instead, the residual is correlated against monthly sunshine hours from the RAF Valley record — a direct radiation-based measurement that is not in the regression. If Thornthwaite underestimates summer ET in a way that β_2 has not absorbed, high-insolation months should carry disproportionately negative residuals, yielding a systematic negative Pearson r .

(iii) **Spatial pattern within clusters.** For C3 (Western-block open dune), the cluster is split by Euclidean distance from the ridge reference point used in S5 at a 1 km threshold (5 forest-adjacent wells, 13 warren-interior wells), and a Mann-Whitney test asks whether forest-adjacent wells have systematically smaller seasonal amplitudes than warren-interior wells — as would be expected if a steady ridge baseflow were flattening the residual at ridge-proximal locations.

S6.3. Result: the ET hypothesis is not supported

The sunshine-hours correlation is consistently negative in sign across every cluster, with network mean $r = -0.063$ and per-cluster means ranging from -0.023 (C4 Forest) to -0.079 (C3 West-Sand). None of the 57 wells exceeds the Bartlett 95% significance threshold of $|r| = 0.15$. The sign is directionally consistent with Thornthwaite slightly underestimating radiation-driven ET, but the magnitude of the bias is below the resolution of the monthly water-level data to detect.

Summer-minus-winter residuals are negative in all clusters except C1 (where they are within noise of zero), with magnitudes of order -0.014 m (C2) to -0.033 m (C4). These values are small relative to the month-to-month variability of the residuals themselves (typical well-level residual standard deviation of order 0.1 m) and should not be read as a strong summer-ET signal. Notably, C4 — the cluster where canopy interception would most plausibly produce Thornthwaite miscalibration — has the weakest sunshine correlation of any cluster, indicating that the cluster-mean β_2 has absorbed forest-specific ET behaviour reasonably well despite the simplicity of the fitted model.

The C3 within-cluster split also returns a null: forest-adjacent C3 wells have slightly larger seasonal amplitudes than warren-interior wells (0.021 m vs 0.019 m; Mann-Whitney $p = 0.88$, alternative “adjacent < interior”). This is the opposite direction from what a steady-ridge-baseflow mechanism would predict, though the difference is not statistically significant.

S6.4. What the residuals actually look like

The per-cluster seasonal climatologies (Figure S6a) reveal a structure that fits none of the candidate mechanisms cleanly: the annual cycle is bimodal, with positive residuals in January–February AND June–July, and negative troughs in April–May and September–October. The sinusoidal fit captures this poorly, which is why the fitted amplitudes appear modest relative to the visible annual range. The phase of the fitted sinusoid shifts systematically later in the year moving from C1 (mid-January) through C2, C3, to C4 (late March). No cluster has a phase in the JJA summer band (Figure S6c).

This bimodal structure is not the signature of unmodelled summer ET, which would produce a single summer trough. It is not the signature of flat steady ridge baseflow, which would produce no annual

cycle at all. And it is not a simple winter-recharge-nonlinearity signature, which would produce a single winter peak. Its most parsimonious interpretation is that a linear lumped-parameter model with time-invariant β coefficients is a rough approximation of a system whose soil-moisture storage operates nonlinearly across the annual cycle — wet-winter saturation increasing recharge efficiency beyond what the cluster-mean β_1 represents, shoulder-season soil drying reducing it below, and the whole cycle modulated by seasonal variation in vegetation water use that a constant β_2 cannot track.

The spatial pattern of seasonal amplitude (Figure S6b) nonetheless shows a coherent structure: the highest amplitudes (≥ 0.04 m) are concentrated in the north-western forest-margin cluster, with CEH14 as the network maximum (0.089 m), followed by CEH34, CEH13, and CEH2 — exactly the same spatial pattern as the Model B intercepts α mapped in Figure 15a. Whatever mechanism is producing the persistent α offset is also producing the elevated seasonal variance. This is consistent with either (a) genuinely larger boundary fluxes at these wells that vary seasonally, or (b) the cluster-mean β values being a worse approximation at ridge-margin wells than at dune-interior wells. Water-level data alone cannot distinguish these.

S6.5. Combined conclusion from S5 and S6

Taken together, Supplementary Notes S5 and S6 have excluded the two leading specific attributions of the water-balance residual:

The residual does not show the distance-dependent lag signature of event-driven ridge transport (S5).

The residual does not show the summer-negative signature of unmodelled Thornthwaite underestimation of atmospheric demand (S6).

The main text of Section 5.3 has been framed accordingly: the model explains 50–70% of month-to-month variance in Δh depending on cluster, and the unexplained portion is real and spatially structured (concentrated at the forest margin) but cannot be uniquely attributed to any single physical mechanism from water-level data alone. The fitted intercept α remains the most defensible summary statistic for the steady-state boundary contribution; the underlying mechanism is most plausibly a combination of steady ridge baseflow, nonlinear soil-moisture storage dynamics not captured by the linear SSM, and minor β coefficient miscalibration, in proportions that cannot be resolved without additional measurement of the kind identified as priority further work in Section 11.

S6.6. Reproducibility

This analysis is implemented in *24_residual_seasonality.py*, which like Scripts 22 and 23 is maintained outside the main pipeline to preserve the reproducibility of the headline results. The script reads the same intermediate data as Script 22 (*01_wells_clean.csv*, *01_climate.csv*, *01_locations.csv*, *02_cluster_stats.csv*) plus the raw RAF Valley climate file for sunshine hours (*RAF_Valley_Climate.csv*). All outputs are written to *24_residual_seasonality/* in the project outputs tree: a per-well climatology table (*24_residual_climatology.csv*), the per-cluster climatology panels (*24_01_climatology_panels_by_cluster.png*), the spatial amplitude map (*24_02_seasonal_amplitude_map.png*), the sunshine-correlation scatter (*24_03_sun_residual_correlation.png*), the per-cluster phase distribution (*24_04_phase_by_cluster.png*), and a plain-text summary of the diagnostic result (*24_05_diagnostic_summary.txt*).